Tropical proof systems

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Good old LP systems

Axioms $x_i \geqslant 0, \;\; 1 - x_i \geqslant 0$.

Cutting Planes (CP) [W.Cook, Coullard, Turán]

$$\frac{f_1 \geqslant 0 \qquad f_2 \geqslant 0}{\alpha_1 f_1 + \alpha_2 f_2 \geqslant 0}\ (\alpha_1, \alpha_2 \geqslant 0), \qquad \frac{\sum_i ca_i x_i - d \geqslant 0}{\sum_i a_i x_i - \lceil d/c \rceil \geqslant 0}\ (c, a_i, d \in \mathbb{Z}).$$

Resolution over Cutting Planes (Res(CP)) [Krajíček]

$$\frac{f_1 \geqslant 0 \ \vee\ \Gamma \qquad f_2 \geqslant 0 \ \vee\ \Gamma}{\alpha_1 f_1 + \alpha_2 f_2 \geqslant 0 \ \vee\ \Gamma}\ (\alpha_1, \alpha_2 \geqslant 0) \qquad \text{(+RES)}$$

$$\frac{\sum_i ca_i x_i - d \geqslant 0 \ \vee\ \Gamma}{\sum_i a_i x_i - \lceil d/c \rceil \geqslant 0 \ \vee\ \Gamma}\ (c, a_i, d \in \mathbb{Z}).$$

$$\frac{\text{—}}{f - 1 \geqslant 0 \ \vee\ -f \geqslant 0}. \qquad \text{(NEG)}$$

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$$\frac{\text{—}}{x - 1 \geqslant 0 \ \vee\ -x \geqslant 0}\ (x \text{ is a variable}). \qquad \text{(NEG-x)}$$

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Facts:

- Exponential lower bound for CP
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Tropical (min-plus) arithmetic

Min-plus arithmetic over $\mathbb{Q}_\infty$:

$$a \oplus b = \min(a, b), \qquad a \odot b = a + b$$

Tropical monomial:

$$m = x_1^{d_1} \odot x_2^{d_2} \odot \ldots \odot x_k^{d_k}$$

Tropical term:

$$t = c \odot m, \quad c \in \mathbb{Q}$$

The empty monomial (term), ∞ plays the role of zero:

$$\infty$$

Tropical polynomial:

$$p = t_1 \oplus t_2 \oplus \ldots \oplus t_m$$

Min-plus inequality:

$$p_1 \leqslant p_2$$

In the usual terms (for linear functions L_i):

$$\min(L_1, \ldots, L_m) \leqslant \min(L'_1, \ldots, L'_s)$$

or

$$L_1 \leqslant L \ \vee \ L_2 \leqslant L \ \vee \ \ldots \ \vee \ L_m \leqslant L$$

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Input system of min-plus inequalities: $f_i \leqslant g_i$.

Min-Plus Nullstellensatz (MP-NS) — complete by [Grigoriev, Podolskii].

Multiply $f_i \leqslant g_i$ by polynomials and take the sum:

$$\bigoplus_i f_i \odot p_i \leqslant \bigoplus_i g_i \odot p_i$$

so that for every monomial (including ∞),
its coefficient on the left is $>$ than on the right.

$1 \odot x \odot y \bigoplus \frac{1}{2} \leqslant 0 \odot x \odot y \bigoplus -\frac{1}{2}$ $\qquad$ $\min(x+y+1, \frac{1}{2}) \leqslant \min(x+y, -\frac{1}{2})$

Min-Plus Polynomial Calculus (MP-PC).

Do it step by step.

- Take the tropical sum $\oplus$.
- Multiply by a term.
- Transitivity of $\leqslant$.
- Thus compose an MP-NS-like inequality.

Tropical resolution rule.

$$\frac{t \oplus f \leqslant 0 \qquad t' \oplus f \leqslant 0}{(t \odot t') \oplus f \leqslant 0}, \text{ where } t, t' \text{ are terms.} \qquad (\odot\text{RES})$$

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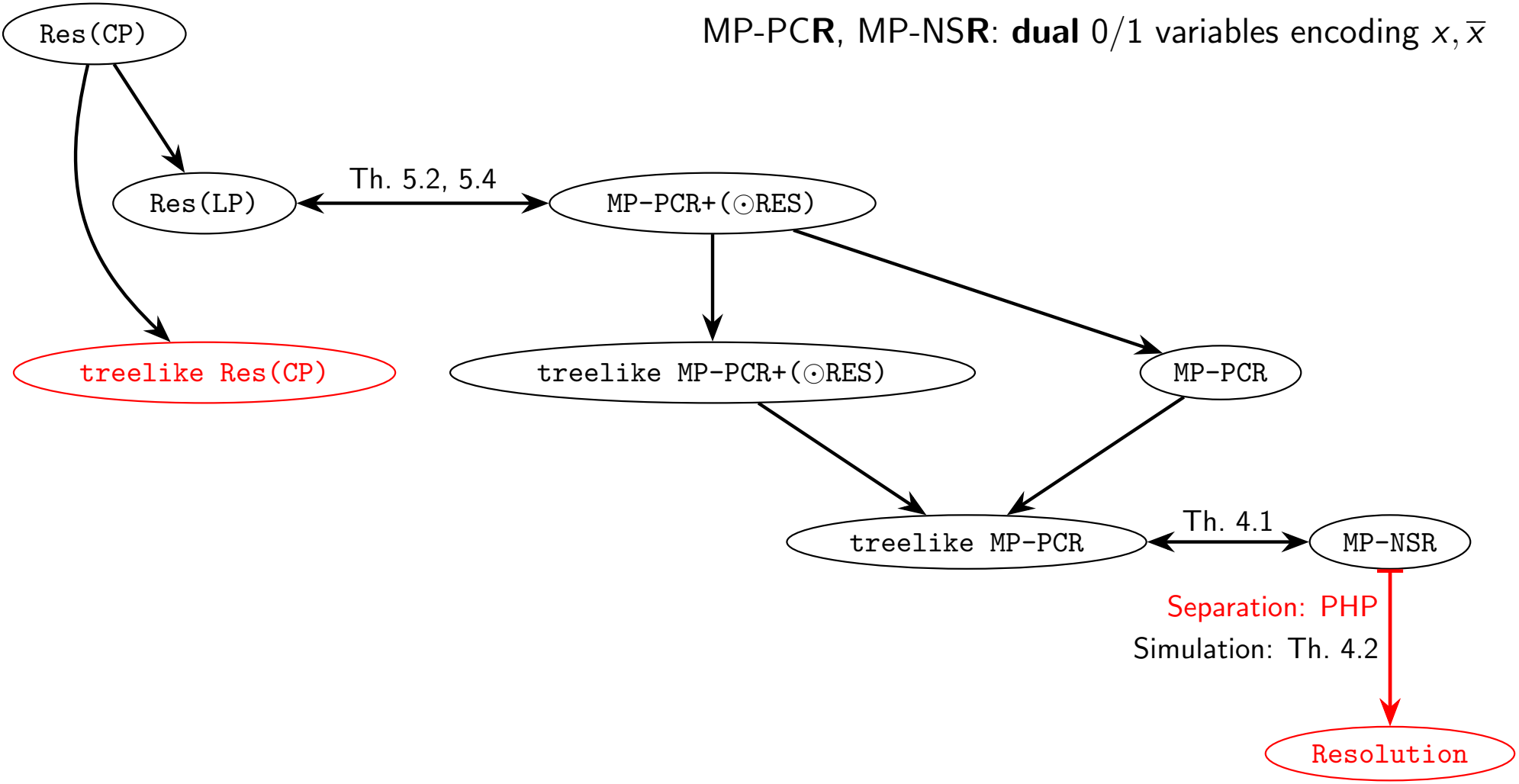
General picture

MP-PC**R**, MP-NS**R**: **dual** 0/1 variables encoding $x, \overline{x}$



Important: there is no equivalence to Res(LP) without (⊙RES), with some evidence.

PHP in treelike Res(LP) and MP-NSR

Treelike Res(LP) proof (motivated by the known treelike CP proof, but avoids rounding).

Lemma

$S_n := x_1 + \ldots + x_n$.
There is a treelike Res(LP) derivation of $S_n \leqslant 1$ from $x_i + x_j \leqslant 1$.

Proof.

- $S_{n-1} \leqslant 1$

  $\vdash S_{n-1} + x_n \leqslant 1 + x_n$.
- $x_i + x_n \leqslant 1$ (for $1 \leqslant i \leqslant n-1$)

  $\vdash S_{n-1} + (n-1)x_n \leqslant n-1$

  $\vdash S_{n-1} + x_n \leqslant 1 + (n-2)\overline{x}_n$.
- Add both inequalities to $x_n \leqslant 0 \lor (n-2)\overline{x}_n \leqslant 0$.

□

It does not use full (+RES), one premise is not a disjunction $\Rightarrow$ Switch to MP-PCR.

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PHP in `treelike Res(LP)` and `MP-NSR`

Switching to tropical Nullstellensatz (or `treelike MP-PCR`).

Lemma

$S_n := x_1 \odot \ldots \odot x_n$.
There is a `treelike MP-PCR` derivation of $S_n \leqslant 1$ from $x_i \odot x_j \leqslant 1$.

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- $x_i \odot x_n \leqslant 1$ (for $1 \leqslant i \leqslant n-1$)

  $\vdash S_{n-1} \odot x_n^{n-1} \leqslant n-1$

  $\vdash S_{n-1} \odot x_n \leqslant 1 \odot \overline{x}_n^{n-2}$.

- Take the tropical sum $S_{n-1} \odot x_n \leqslant (1 \odot x_n) \bigoplus (1 \odot \overline{x}_n^{n-2})$
  and substitute $(1 \odot x_n) \bigoplus (1 \odot \overline{x}_n^{n-2}) \leqslant 1$ □

Now it's a treelike `MP-PCR` refutation.
Some technicalities are needed for $x \oplus \overline{x}^{n-2} \leqslant 0$ and to make the proof static.

Daglike Resolution in MP-NSR

Translate $A = l_1 \vee \ldots \vee l_k$ into $[A] = 0 \odot l_1 \odot \ldots \odot l_k$.

Input clauses

$$1 \leqslant 0 \odot [C_i]. \tag{1}$$

At step $i = 1, 2, \ldots, s$, let $c_i = 1 - 1/(i+1)$.

Resolution

$$\frac{A \vee x \quad A \vee \neg x}{A},$$

becomes

$$c_i \odot x \odot [A] \bigoplus c_i \odot \overline{x} \odot [A] \leqslant c_i \odot [A]. \tag{2}$$

Weakening

$$\frac{A}{A \vee \ell},$$

becomes

$$c_i \odot [A] \leqslant c_i \odot [A] \odot [\ell]. \tag{3}$$

Take the tropical sum of all these (1), (2), (3). They combine nicely!

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Encoding of Boolean logic

- Dual encoding: axioms

$$x \odot \overline{x} = 1, \qquad x \oplus \overline{x} = 0$$

- A Boolean clause:

$$\overline{x}_1 \oplus \overline{x}_2 \oplus \ldots \oplus \overline{x}_n \leqslant 0$$

- Or, like in CP:

$$1 \leqslant x_1 \odot x_2 \ldots \odot x_n$$

- Economic encoding:

$$x^2 \oplus 1 = x \qquad \min(2x, 1) = x$$

- Also $(0, \infty)$-encoding: turns MP-PC into Res(∞)

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Lower bounds

- Linear degree of MP-NSE refutations of the knapsack $\bigoplus_{1 \leqslant i \leqslant n} x_i = r < 0$
- Exponential size bound:

  MP-NSR refutation size for the binomial $x^{\odot k} = c$ is greater than k
- Non-integer coefficients are needed in MP-NSE and MP-NSR
- Tropical resolution ($\odot$RES) is not derivable in MP-PCR
- $0 \leqslant x$ is not derivable in MP-PCE (yet it is complete!)

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Further research

Two sources of non-treelikeness

- Integer multiplication by a big constant: treelike in LP, daglike in tropics.
- Transitivity of min-plus inequalities: treelike in tropics, daglike in LP.

Tropical resolution. Separate or simulate `Res(LP)` in `MP-PCR` without ($\odot$`RES`). Unary coefficients?

`MP-NSR` vs `CP`.

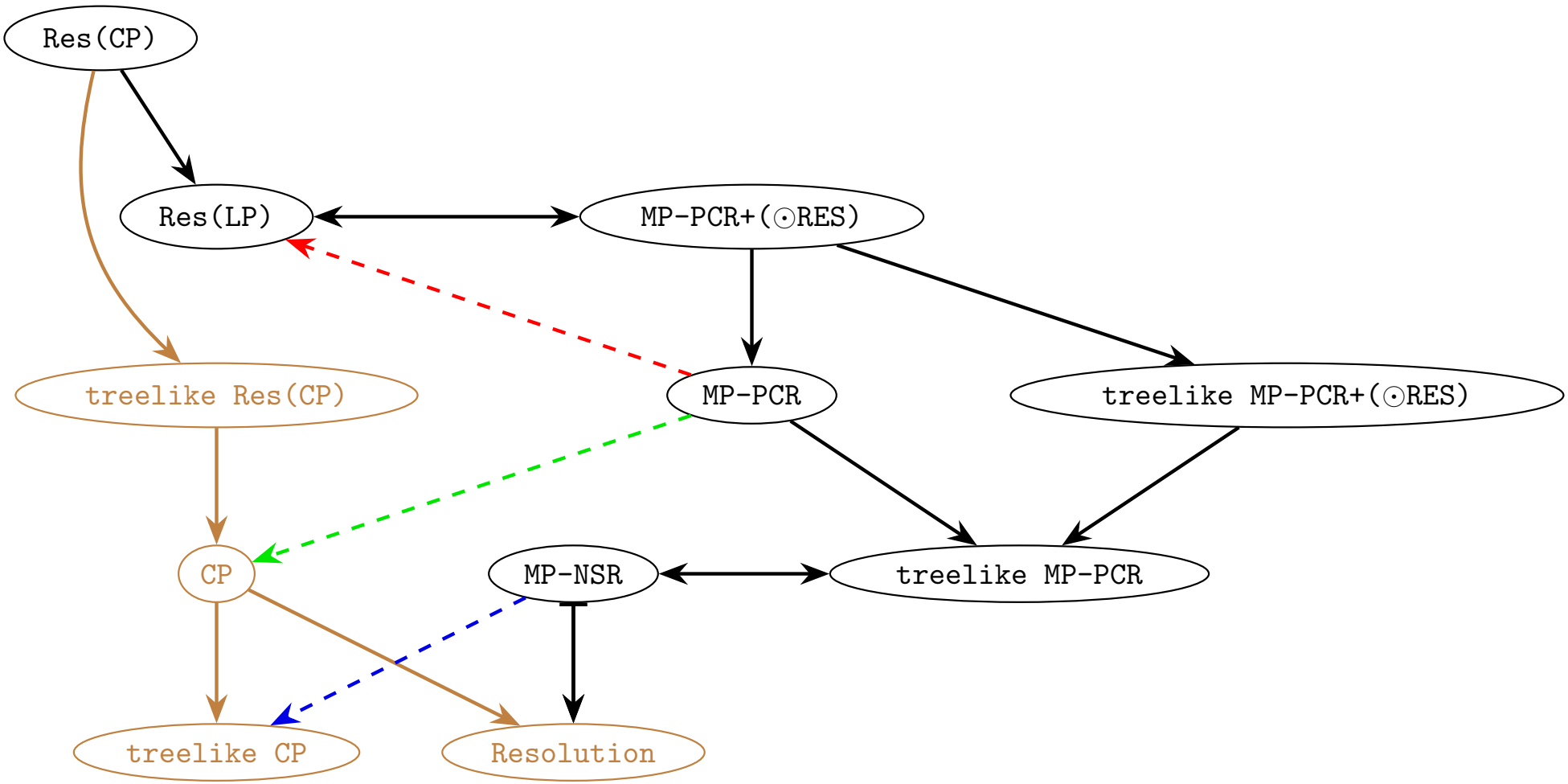
- Relations between `MP-NSR` and `CP` are unclear, both for unary and binary coefficients, and even for `treelike CP`.
- Does adding the integer negation (as in `Res(CP)`) to `MP-NSR` allow it to simulate at least `treelike CP`?

Economic encoding. $x^2 \oplus 1 = x$

- Simulate `Resolution` in `MP-NSE`.
- Show exponential lower bounds for `MP-NSR` or `MP-NSE` for CNFs (or small degrees).

(There are more open questions...)

General picture



Pictures are inspired by Proof Complexity Zoo by Marc Vinyals